



ROOT CAUSE UNLOCKED · CONDITION KIT

The Chemical Sensitivity Root Cause *Kit*

Every test I order, why I order it, how to read it, and what actually changes when the number comes back

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

The Chemical Sensitivity Root Cause Kit

Every test I order, why I order it, how to read it, and what actually changes when the number comes back

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N-acetylcysteine, 600 mg once daily, titrating to 600 mg two or three times daily as tolerated

Liposomal or acetyl-glutathione, 500 to 1,000 mg per day

Selenium, 100 to 200 mcg per day as selenomethionine

Alpha-lipoic acid, 100 mg per day titrating to 300 to 600 mg per day

Milk thistle, silymarin 150 to 300 mg standardized, twice daily

Thiamine, and why it sits upstream of everything above

Magnesium glycinate, 200 to 400 mg elemental per day

Vitamin C, 250 to 500 mg per day total

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Glycine, 3 g per day

Taurine, 500 to 1,000 mg twice daily

Zinc, 15 to 30 mg per day, with 1 to 2 mg copper

Quercetin, 200 mg per day of a bioavailable form

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. Nothing in here diagnoses multiple chemical sensitivity or cures it.

Same-day medical attention, not a lab plan. Difficulty breathing, chest pain, swelling of the lips or tongue, fainting, confusion, a first seizure, or a reaction that keeps escalating after you leave the exposure. Those are emergencies. No panel result changes that, and none should delay that call.

Nothing in this guide is a reason to stop, skip, or reduce a prescription. If you take an inhaler, an antihistamine, a thyroid medication, an antidepressant, or anything else, every decision about that prescription belongs with the physician who wrote it. What is here sits alongside that care, never instead of it.

One caution specific to this condition, and I want it near the top. People with chemical sensitivity often react to supplements too, sometimes to the ones intended to help. The published example is sobering and worth stating precisely: in an Italian case series of chemically sensitive patients, seven of the group received reduced glutathione by continuous intravenous infusion, and five of those seven had adverse reactions (PMID 24367721). That was an intravenous infusion, not a capsule, so it does not transfer directly to oral dosing. It does tell you why I move slowly, one agent at a time, and why I would rather have your labs in hand before I ask you to swallow anything.

What you are holding, stated plainly

You are about to spend real money on testing, so here is the honest account of what this workup is before you read a single result.

Some of what I order is ordinary, validated medicine, such as a blood count, a metabolic panel, thyroid tests, and iron studies. Those are well established and I read them the way any clinician would. The rest of it, the functional and environmental panels, sits in a different category. Two independent research groups have measured the glutathione system in people carrying a clinical diagnosis of multiple chemical sensitivity and found it depleted (PMID 40943945, PMID 20430047), and the biochemistry of which nutrients that system runs on is established in humans. What does not exist is a trial showing that testing people this way, and then correcting what the tests find, makes chemically sensitive people function better than not doing it. That trial has not been run, and nobody is lining up to fund it, because almost everything it would lead to is unpatentable.

So the evidence tier here, stated once: established laboratory measurement and established human biochemistry, no outcome trial in this population, held to an endpoint. The endpoint is your own symptom and exposure log compared against your own baseline, on a retest. If a marker moves and you do not feel or function better, the marker was not the story, and I would rather find that out in three months than defend it for three years.

There is one more thing this framing buys you. Low glutathione in a chemically sensitive person may be a consequence of the illness as easily as a cause of it. I do not know which, and neither does anyone else yet. That is a genuine hole, and my answer to it is to measure, correct what is correctable, recheck, and let your function settle the question, rather than to argue.

How I read a test, and the words I will and will not use

Four rules govern every entry below. They are the difference between a workup and a sales pitch.

Urinary panels document recent exposure. They do not measure what is stored in your tissues, and they do not tell us where the exposure came from. A urine result is a snapshot of what your kidneys were clearing during the collection window.

Functional markers are suggestive rather than diagnostic. They generate a short list of things worth investigating. They do not name your condition, and I will not let one drive a decision on its own.

“Consistent with” is almost always the right phrase. “Confirms” almost never is. There are real exceptions, and they matter: an unprovoked blood lead level genuinely confirms lead toxicity, because that test was validated for exactly that use. Where a test has been validated for the question being asked, I will say so and use the stronger word. Where it has not, I will not borrow the authority.

A test earns its place by changing something. Before I order anything, I should be able to tell you what I would do differently for a high result, a low result, and a normal one. If I cannot answer that, the test is not ready to be ordered, and you should hold any clinician to that standard, including me.

The thing no lab result outranks

Reducing your exposures is the foundation of managing this condition, and the reason deserves a plain account. It has never been tested against a control group, and no study has ever compared it head to head against supplementation (PMID 34769816). I recommend it first because it is low risk and it directly removes the thing that is making you sick, not because a trial showed it beats anything else. No result in this guide changes that ordering. A perfect panel and an unchanged environment is a bad trade.

Part 1: The baseline panel

This is the ordinary medicine, and I would not skip it for anyone. Finding one genuinely treatable thing here is worth more than months of targeted supplements aimed at a driver that was never the driver.

Complete blood count with differential

What it measures. Red cells, white cells, and platelets, with the differential breaking the white cells into their types.

Why I order it. Anemia, infection, and marrow problems all produce fatigue, headache, breathlessness, and cognitive fog, which is the same list chemical sensitivity produces. I want those off the table.

How to read it. Read it as ordinary medicine, not as a chemical sensitivity marker. One case-control study did compare blood counts between chemically sensitive people and controls and found the condition associated with a slightly lower lymphocyte count and a slightly higher mean cell hemoglobin concentration (PMID 15347780). Those are population-level associations too small to identify anyone, and the same paper concluded its findings were inconsistent with the proposal that chemical sensitivity comes with vitamin deficiency or thyroid dysfunction. I mention it so that nobody sells you a lymphocyte count as proof of anything.

What changes. A low hemoglobin sends me to iron studies before anything else. An abnormal white count or differential gets worked up on its own terms with your physician, and the rest of this plan waits.

Comprehensive metabolic panel

What it measures. Kidney function, liver enzymes, electrolytes, blood glucose, protein, and albumin.

Why I order it. Two reasons. First, kidney and liver function are the organs doing the actual clearance work, and I need to know their condition before I add anything to your load. Second, several markers later in this guide are read differently when kidney function is impaired.

How to read it. Read the kidney and liver numbers first. The case-control work in chemical sensitivity found a positive association with alanine aminotransferase, one of the liver enzymes (PMID 15347780), which is another population-level signal rather than something that identifies an individual. The glucose on this panel is a single fasting value and a single value is a weak read of fuel stability.

What changes. Impaired kidney function changes how I read methylmalonic acid and homocysteine, both of which rise when clearance falls, and it makes me more conservative with dosing across the board. A raised liver enzyme gets explained before I add supplements, not after. If the fasting glucose is drifting up, or your symptoms track hard with meals and fasting, the honest next step is the fuller blood sugar workup, which means hemoglobin A1c and a fasting insulin alongside the glucose, because insulin rises years before glucose.

RBC magnesium

What it measures. Magnesium inside your red blood cells rather than floating in serum.

Why I order it. Magnesium is a cofactor across energy production and nervous system regulation, and my protocol uses magnesium glycinate, so I want a starting number to compare against.

How to read it. With humility. Magnesium status is genuinely hard to measure, which is exactly what the systematic review of the available assessment methods set out to examine (PMID 22064327), and I do not treat any single number as a definitive readout of your total body magnesium. There is a second honesty here that most programs skip. The largest study to measure minerals inside red blood cells in chemical sensitivity, 408 women meeting validated criteria for the condition, found no statistically significant differences from controls in any mineral it tested, magnesium included, and its authors concluded plainly that these measurements do not provide useful diagnostic markers for the condition (PMID 17046989).

What changes. A genuinely low result means I correct it and recheck. A normal result does not mean magnesium is irrelevant to you, and it does not mean I will not use it. What it does mean is that I am no longer treating a documented deficiency, and I say so out loud rather than pretending the number justified the capsule.

Zinc and copper with ceruloplasmin

What it measures. Serum zinc, serum copper, and ceruloplasmin, the protein that carries most of your circulating copper.

Why I order it. Zinc is required across immune function and wound healing, and zinc supplementation drives copper down over time. My protocol pairs zinc with a small amount of copper for exactly that reason, and I would rather know your starting position than assume it.

How to read it. Zinc deficiency is often described as one of the most consistent findings in chemical sensitivity. The largest study that actually measured it found no significant difference (PMID 17046989). Copper trended lower in that study too, as did four other minerals, all without significance and without a dose-response relationship. Serum zinc has its own measurement limitations, catalogued in the systematic review of zinc assessment methods (PMID 38917458), including the fact that it falls during inflammation independently of your actual zinc stores.

What changes. A low zinc with a normal inflammatory marker is a real deficiency and I correct it. With a high hs-CRP, the same low zinc is a number to repeat when the inflammation settles rather than a diagnosis. A copper or ceruloplasmin drifting down on zinc is my signal to adjust the copper, and it is the single reason I recheck this pair rather than ordering it once.

25-OH vitamin D

What it measures. 25-hydroxyvitamin D, the storage form of vitamin D your body draws on. This is the correct analyte for assessing vitamin D status (PMID 21646368), and I mention that because the shelf is full of tests that measure the wrong molecule.

Why I order it, and the position that runs against me. You deserve both halves. The 2024 Endocrine Society guideline advises against routine 25-hydroxyvitamin D screening and states that no clear evidence defines an optimal level (PMID 38828931). That is a real, current, well-made guideline and I am not going to hide it from you.

Here is my position and the distinction I think matters. A guideline about screening asymptomatic populations is answering a different question than case-finding in a symptomatic person who is already in my office describing fatigue, pain, and poor recovery. I test in the second situation, not the first. My functional target is 40 to 80 ng/mL, and I am telling you plainly that this is my clinical target, not a validated diagnostic threshold, and not a number the trials established.

How to read it. Correct a genuinely low level. Do not chase the top of the range. When a three year randomized trial gave healthy adults high doses that landed them in the upper part of that band, the result was not extra benefit, it was slightly lower forearm bone density in the higher dose arms (PMID 31454046), and a very large trial at 2000 IU daily was null for its main outcomes (PMID 30415629).

What changes. A low result gets corrected with a dose matched to how low it is, taken with food, and rechecked. A result already inside the target changes nothing, and specifically does not earn a higher dose.

Ferritin and the iron panel

What it measures. Ferritin estimates iron stores. The panel adds serum iron, total iron binding capacity, and transferrin saturation.

Why I order it. Iron deficiency without anemia is common and it produces fatigue, breathlessness, headache, and poor exercise tolerance before the hemoglobin ever falls. It is one of the most fixable things on this list.

How to read it. Ferritin is an acute-phase reactant, meaning it rises with inflammation regardless of your iron stores. It has to be read next to hs-CRP. A ferritin that looks acceptable during an inflammatory episode can be hiding genuinely depleted iron, which is why I read the saturation and binding capacity alongside it rather than the ferritin alone. My functional target is 50 to 150 ng/mL.

What changes. Low iron gets corrected and the cause gets found, because in adults the cause matters as much as the correction. High ferritin with a high saturation is a different conversation entirely and belongs with your physician, not with a supplement plan.

Homocysteine, read correctly, including when it is low

What it measures. An amino acid that sits at a junction in the methylation cycle. It rises when B12, folate, or B6 run short, and with thyroid or kidney trouble.

Why I order it. Because it is one of the most consequential numbers in this panel and it is routinely waved off. An elevated homocysteine is telling you the methylation machinery your nervous system and your detoxification pathways depend on is running short somewhere findable. My functional target is 5 to 7 umol/L, which reflects how seriously I take it. Note that it is a band with a floor and not a “the lower the better” instruction, and the next two paragraphs are about why.

How to read it when it is high. Not by chasing the number with blind supplementation. Find which input is short, B12, folate, B6, thyroid, or kidney function, correct that one, and let homocysteine confirm the fix on the retest.

How to read it when it is low, which is where this guide has to be careful. Lower is not better. My target has a floor of 5 for a reason: a homocysteine driven below that band is where I start watching for overmethylation and B vitamin intolerance, and it is a signal to slow down with methyl donors rather than to congratulate ourselves on the number. That is my clinical position, stated as one, and you should know exactly what it is and is not made of. No published reference interval, cutoff, or validation study assigns an “overmethylation” meaning to a low homocysteine, and this test does not measure methylation status inside your cells, which is governed by the ratio of two other molecules that this panel does not touch. The established biochemistry actually runs the other direction: a low value most often reflects good B vitamin status, or brisk clearance of homocysteine down the transsulfuration pathway toward cysteine (PMID 20301697). There is one intriguing data point that is not an explanation: the case-control study of chemically sensitive patients found the condition associated with lower total plasma homocysteine, not higher (PMID 15347780). That study drew no methylation conclusion from it, and neither will I.

What changes. High homocysteine sends me hunting for the short input and I recheck after correcting it. Low homocysteine does not make me stop methyl donors, because there is no evidence that it should. It makes me introduce them one at a time at a fraction of the target dose, and judge by how you feel over the following week rather than by the number. If methyl donors reliably flare you, I switch to hydroxocobalamin and folinic acid, and I will tell you honestly that this is clinical convention rather than evidence. Nobody has published a definition of an “overmethylator” or a test that identifies one, and no study has compared hydroxocobalamin against methylcobalamin in people who react badly to B vitamins. Folinic acid is a real and well studied folate form, but the condition it is proven for is a specific antibody-mediated folate transport problem, which is not the same thing as reacting to methyl donors.

hs-CRP

What it measures. High-sensitivity C-reactive protein, a general inflammatory marker.

Why I order it. Because it is inexpensive and because several other markers on this panel cannot be read correctly without it. Ferritin and serum zinc both move with inflammation.

How to read it. As a nonspecific screening marker. It says an inflammatory load exists somewhere. It does not say where it came from, and it never confirms anything. My functional target is under 0.5 mg/L. I have no study showing that this marker distinguishes chemical sensitivity from anything else, and I do not use it that way. A single high reading after an illness, an injury, or a hard training week means very little. Repeat it when you are well.

What changes. A high result makes me reread the ferritin and zinc rather than trust them. It also makes me look harder at the obvious sources of inflammatory load, including the exposures themselves, before I add anything. It does not by itself start a supplement.

TSH, free T4, and free T3

What it measures. The pituitary signal and the two circulating thyroid hormones.

Why I order it. Thyroid disease produces fatigue, cold intolerance, cognitive fog, constipation, and low mood, and it is common, treatable, and easy to miss.

How to read it. As a thyroid workup, not as a chemical sensitivity marker. The case-control study that looked found no association between chemical sensitivity and thyroid function (PMID 15347780). My functional target for TSH is 1.0 to 2.5 mIU/L. For free T4 and free T3 I publish no functional target, because I do not have one that holds up across the whole shelf, and I would rather tell you that than print a number I cannot defend. Your laboratory's reference range and your clinician's read govern those two.

What changes. A genuinely abnormal thyroid result goes to the physician who would manage it, and correcting it is worth doing on its own terms whatever else is going on. Thyroid medication decisions are never mine to make, and I route them to your prescriber every time.

AM cortisol

What it measures. Your cortisol level in the morning, when it should be near its daily peak.

Why I order it. As a screen for genuine adrenal insufficiency, which is a real disease that presents as fatigue, low blood pressure, and poor stress tolerance, and which nobody should miss while attributing the picture to an environmental illness.

How to read it, and what it is not. I want to hold two things apart here, because the market blurs them deliberately. “Adrenal fatigue” as a diagnosis is not recognized, no panel confirms it (PMID 27557747), and the glandular and adaptogen protocols sold on that promise stay off my list. On top of that, the specific claim that adrenal disorders show up more often in chemically sensitive people does not survive contact with the data: repeated low level chemical exposure does shift stress hormone signaling in rodents (PMID 11306018, PMID 15196973), but both human studies that actually measured cortisol in chemically sensitive subjects, one of them including a dexamethasone suppression test, found no difference from controls (PMID 16680381, PMID 14626900). So I do not tell you your adrenals are the problem.

What I do use, and never remove from a workup, is four-point salivary cortisol across a day when the morning number does not match your story. A flat and low pattern and a wired and high pattern both present as exhaustion and call for opposite approaches, and a single morning draw cannot tell them apart. I read the shape of that curve as a clinically useful pattern, not as a validated diagnostic threshold, and I publish no functional target for it.

What changes. A morning cortisol suggesting true adrenal insufficiency goes straight to a physician, urgently, and everything else waits. A normal one closes that question. The diurnal curve changes the timing and pacing of the plan, particularly around sleep, meals, and when in the day I ask you to attempt anything demanding.

Prolactin

What it measures. A pituitary hormone.

Why I order it, and the honest evidence position. There is no published study linking chemical sensitivity to high prolactin. I searched for one. The single study that has ever measured prolactin in chemically sensitive subjects found a trend toward lower prolactin, the opposite direction, and it did not reach statistical significance (PMID 14626900). I am not going to imply otherwise.

How to read it. As a general medical test, which is exactly why it stays on my list. A raised prolactin has real explanations worth finding: certain medications raise it, and so do pituitary and thyroid conditions that present with headache, menstrual irregularity, low libido, or breast discharge. Those symptoms overlap heavily with what people bring me, and I would rather find a treatable cause than name a syndrome.

What changes. A raised result is a referral, starting with a medication review and then to the physician who would evaluate it. A normal result closes a differential and changes nothing else.

Part 2: The functional panel

This is where the caution in the framing section earns its keep. Every test below is a real laboratory measurement. None of them is validated to diagnose chemical sensitivity, and I read all of them as suggestive rather than diagnostic.

Organic acids, and pyroglutamate specifically

What it measures. Urinary metabolites left over from cellular metabolism, microbial activity, and nutrient handling. I am ordering it for two analytes in particular.

Why I order it. Pyroglutamate, also called 5-oxoproline, is on the panel as a proposed read on glutathione turnover, and methylmalonic acid is on it as a functional read on whether your B12 is working inside cells.

How to read it, honestly. These two analytes are not equal. **Methylmalonic acid is validated.** It is a more sensitive marker of tissue level B12 sufficiency than serum B12 alone (PMID 8154512, PMID 2339684, PMID 23301732), and checking it before committing anyone to B12 therapy is defensible practice. It also rises when kidney function is impaired, which is why it is read next to the metabolic panel.

Pyroglutamate is a reasoned inference, not a validated glutathione test. The mechanistic basis holds up. Urinary 5-oxoproline excretion rose in patients recovering from severe malnutrition and fell when glycine was supplemented, which is what you would expect if glycine, one of the three building blocks of glutathione, had become the rate-limiting supply (PMID 8914954). That was a severe malnutrition population, not a chemically sensitive one, and no study has validated this marker as a glutathione status test in anybody who looks like you. The panel as a whole has a defined professional scope, and it is not this one: the technical standard for organic acid analysis scopes it to the biochemical diagnosis of inherited metabolic disease (PMID 29543224, PMID 33026349).

What changes. A high methylmalonic acid with a low or low-normal B12 is a functional B12 deficiency and I correct it, which a B12-only test would have waved through. A high pyroglutamate makes me look specifically at glycine, which is already in my protocol at 3 g/day, and it makes me want a direct glutathione measurement rather than acting on a urine marker alone. It does not by itself start a glutathione protocol.

Plasma or RBC amino acids

What it measures. The circulating or intracellular levels of individual amino acids, including glycine, cysteine, glutamine, and taurine.

Why I order it. Glutathione is a tripeptide built from glutamate, cysteine, and glycine, and the enzyme that joins the first two is the rate-limiting step in making it (PMID 18601945). This panel tells me whether the raw material is short before I spend your money on the downstream agents.

How to read it. Amino acid profiling is a real analytical method that can generate useful clinical indices and support nutritional decisions (PMID 19057187). What it has not been validated to do is tell us that any particular pattern of cysteine, glycine, or taurine predicts your glutathione capacity or your reactivity. Suggestive, not diagnostic.

What changes. A low glycine supports the glycine already in my protocol and gives me a number to recheck against. A low taurine does the same for taurine. Nothing on this panel changes the pace of introduction, which stays one agent every three to five days at a quarter to half the target dose regardless of how the amino acids look.

Whole blood or RBC glutathione, with the GSH to GSSG ratio

What it measures. Reduced glutathione and its oxidized form, and the ratio between them, which estimates the redox state inside your cells.

Why I order it. This is the single most defensible functional test in the workup for this condition, and it is the one that anchors the whole glutathione arm of the protocol. Two independent research groups have measured the glutathione system in patients carrying a clinical diagnosis of multiple chemical sensitivity and found it depleted (PMID 40943945, PMID 20430047), and a third line of work found redox alterations tracking a superoxide dismutase gene variant in this population (PMID 34667598).

How to read it. The specimen matters and it is not negotiable: glutathione sits at far lower concentrations in plasma than inside cells, which is why this has to be whole blood or red cells rather than plasma, and why the ratio rather than the raw number is what estimates cellular redox state (PMID 28807817). Read it as suggestive rather than diagnostic. It does not diagnose chemical sensitivity, and a depleted result does not tell us whether the depletion is a cause of your illness or a consequence of it.

What changes. A depleted result is the strongest single argument for the glutathione arm of the plan, and it gives us something to retest, which matters more than the first number. There is direct evidence that this marker moves and can be tracked: a six month randomized trial of plain oral glutathione raised whole blood glutathione by 17 percent and red cell glutathione by 29 percent in healthy adults (PMID 24791752).

A depleted result does not license going fast, and here is where I have to be specific about routes. Nebulized and intranasal glutathione genuinely work as delivery methods in humans, which is well established: nebulized glutathione raises glutathione in the airway lining three to four fold, and intranasal glutathione measurably raises it in the brain (PMID 14726422, PMID 28725693). The problem is the population. In a randomized crossover trial in patients with mild asthma, nebulized glutathione caused a 19 percent drop in lung function with coughing and breathlessness, attributed to sulfite formation (PMID 9279219). If you have reactive airways, that is a reason to avoid the nebulized route, not a reason to choose it. And the larger trials of these routes in cystic fibrosis and Parkinson's disease showed delivery without clinical benefit (PMID 23631796, PMID 28436395).

Whole blood vitamin B1, for thiamine status

What it measures. Thiamine diphosphate, the active coenzyme form of vitamin B1, measured inside your blood cells rather than floating in plasma. The test to order is **LabCorp test 121186, Vitamin B1 Whole Blood**, the Mayo equivalent of which reports the same analyte. It must be collected as **whole blood, not plasma**. The coenzyme sits inside the red cells, and a plasma tube discards most of what the test is for.

Why I order it, and why thiamine sits upstream of everything else in the glutathione plan. Transketolase is the thiamine-dependent enzyme in the non-oxidative branch of the pentose phosphate pathway, and that pathway is a major supplier of NADPH, which is the reducing power that regenerates used glutathione back into its active form (PMID 34685573). If thiamine is short, the glutathione recycling machinery is running on a restricted power supply, and adding glutathione precursors downstream is pushing a rope. This is why thiamine comes before the glutathione section in my sequence rather than after it.

How to read it. By the whole blood level, not by a serum thiamine, which is a poor reflection of functional status. The reference range is **70 to 180 nmol/L of thiamine diphosphate, and a result below 70 suggests deficiency** (PMID 34685573).

The honest limit of this test, which you need before you read your result. A normal blood level does not exclude a functional deficiency. The number tells you how much coenzyme is present, not whether the enzyme that depends on it is keeping up. That distinction is exactly what the older literature was measuring with the erythrocyte transketolase activation test, which added thiamine to the sample and watched how much the enzyme sped up. You will see that assay named in the papers, and it is worth knowing it is not commercially orderable in the United States, so it is not something you can hand a requisition for. What I do about that gap is read the whole blood level next to your symptoms and your response to a trial, rather than treating the number as the whole answer.

The evidence tier for this one, plainly. The biochemistry is established in humans. There is no clinical trial of thiamine in multiple chemical sensitivity. None. Instead of shrugging at that gap, I test rather than assume, I correct a documented deficiency, and I recheck rather than treating indefinitely on a theory.

What changes. A whole blood B1 below 70 nmol/L moves thiamine to the front of the sequence, at benfotiamine 150 mg twice daily for 4 to 6 weeks, and it gets rechecked. A normal result does not by itself close the question, because of the limit above. It does mean I stop treating thiamine as the obvious bottleneck and move on to the glutathione and conjugation cofactors. The same logic applies to riboflavin, whose functional status is measured through a closely related red cell enzyme assay: riboflavin becomes FAD, and FAD is the required cofactor for glutathione reductase, the enzyme that recycles used glutathione, and for the flavoprotein that powers liver Phase I enzymes (PMID 12791609, PMID 37603429, PMID 19102813, PMID 22982532). That mechanism is established. The human trials that exist used about 1.6 mg of riboflavin a day, so any higher dose you see on a label is practitioner convention rather than a studied amount, and I will say so rather than imply trial backing.

Urinary mycotoxin panel

What it measures. Mycotoxin metabolites cleared in your urine during the collection window.

The published position, printed first. Current German guideline work on indoor mold exposure states that blood or urine tests for mold components or metabolites are not indicated in the medical evaluation of mold exposure (PMID 38381662). That is the guideline position and you should have it before you spend anything.

The rest of the honest picture. The biology holds up. Mycotoxins are absorbed, conjugated, and cleared renally, and validated urinary exposure biomarkers exist within exposure science for aflatoxin, deoxynivalenol, ochratoxin A, and fumonisin (PMID 33350155). The problem is clinical validity rather than chemistry. The developers of those biomarkers themselves flag short half-lives and interpretation pitfalls (PMID 33924868). The clinical study most often cited for these panels compared patients against a non-concurrent, unblinded historical control group and drew two published methodological critiques (PMID 23580077, PMID 27827982, PMID 27827984, PMID 31608429). And dietary mycotoxin exposure is close to universal, which means a positive result cannot localize the source to your building or attribute your symptoms to it.

Where I stand, labeled as mine. I order this panel. That is my clinical practice, not guideline-concordant care, and I am not going to dress it up as the latter. I order it because when the exposure history and the building point the same direction, I would rather have the measurement in the record than not, and because it occasionally redirects a plan that was heading somewhere else entirely.

How to read it. As documentation of recent exposure. Not as a diagnosis, not as a measure of what is stored in your tissues, and never as the sole reason to start treatment.

What changes. If your home or workplace has dampness or visible mold, the first action is correcting the moisture and the building damage, and that is true whether the panel comes back positive, negative, or ambiguous. A positive result never by itself starts a binder. Binders enter only where the picture holds up across the exposure history, the building, and the panel together, and only after your bowels are moving reliably every day, because a binder in a constipated gut is a bad idea. When binders do go in, they go 1 to 2 hours away from food, other supplements, and medication, at fractional doses.

Heavy metals

What it measures. Blood testing measures what is circulating now, which reflects recent or ongoing exposure. Unprovoked urine measures what your kidneys are clearing.

Why I order it. Only when the exposure history supports it. Occupation, hobbies, home age, water source, imported cookware or cosmetics, and specific documented incidents are what put this on the list, not the symptom picture alone.

How to read it. This is the one place in the guide where the stronger verb is correct, because the test was validated for the question: an unprovoked blood lead level confirms lead toxicity. That is established diagnostics, not a functional panel, and I will use the word “confirms” for it and almost nowhere else.

What I will not order, and why. I do not use provoked urine testing, which administers a chelating agent before collecting the sample. Two reasons, both concrete. It does not do what it claims: in a prospective cohort, chelator-provoked urine results failed to predict heavy metal poisoning (PMID 34184587). And the agents involved carry real harm, including deaths from hypocalcemia following administration of the wrong edetate formulation (PMID 16882789). Provoked results also cannot be compared against standard reference ranges, which is precisely how they lead directly to chelation nobody needed. I do not use chelation as a general detoxification strategy, and I would be very cautious of any practitioner who does.

What changes. A genuine documented poisoning is a medical toxicology referral, not a supplement plan, and everything else waits. A negative result closes the question and I stop spending your money there.

One related point belongs here, because my protocol includes alpha lipoic acid and you will find the internet worried about it. The concern that lipoic acid mobilizes mercury has a real mechanism behind it, and it is narrower than it is usually described. In rats, intravenous lipoic acid increased biliary excretion of inorganic mercury substantially, and it decreased biliary excretion of methylmercury, the form most people worry about from fish (PMID 1585376). The movement that was demonstrated was into bile for elimination, not redistribution into the brain. No human study has measured this, and those animal doses were intravenous and far higher per kilogram than an oral capsule. I am not going to overstate a brain redistribution risk that was never demonstrated.

Genetic variants (GSTM1, GSTP1, MTHFR, COMT, SULT)

What it measures. Inherited variation in enzymes involved in conjugation, methylation, and sulfation.

Why I order it. As supportive context that occasionally changes which nutrient I reach for first. Never as a diagnosis, and never as an explanation for your reactivity.

How to read each one, because they are genuinely not equal.

GSTM1 and GSTT1 are true deletions that remove a working detoxification enzyme in a large fraction of people. One study of 521 adults found these deletions more common in those reporting chemical sensitivity (PMID 17291352). Three later studies in formally diagnosed cohorts found no such difference (PMID 20430047, PMID 22205546, PMID 23967348, PMID 15256524). **GSTP1 has no deletion variant at all**, only coding changes, so any report describing a GSTP1 “null” is describing something that does not exist. Taken together, these genes cannot be said to explain anyone’s reactivity, and I will not present them to you as an explanation.

MTHFR C677T is the one variant here with genuine clinical traction. The variant destabilizes the enzyme’s FAD binding, and randomized genotype-targeted trials show that adults with the 677TT genotype respond to riboflavin when other genotypes do not, at 1.6 mg a day (PMID 7647779, PMID 10201405, PMID 23608654, PMID 22277556), with riboflavin measurably shifting DNA methylation specifically in that genotype (PMID 32334045). Genotype changing the response to a B vitamin is a proven human fact here, not a hypothesis.

COMT Val158Met is a well established enzyme kinetics variant, and the enzyme does consume methyl groups (PMID 8807664, PMID 15457404). What has never been tested is whether COMT genotype predicts who reacts badly to methyl donor supplements. That is a reasonable hypothesis, not a finding.

SULT is where the sulfation hypothesis in chemical sensitivity comes from, and it is weaker than it is usually presented. The originating probe drug work found impaired sulfation of phenolic compounds in a portion of people with environmental intolerance or chronic disease (PMID 8711748). That is a single line of older work, it does not establish that a genetic variant explains your symptoms, and I have found no trial testing whether acting on it changes anything.

What changes. A 677TT result changes which B vitamin I reach for first and gives riboflavin a real reason to be early in the sequence. COMT and SULT results change nothing on their own; at most they make me slower and more careful introducing methyl donors and sulfur-heavy agents, which is how I introduce them anyway. GST results change nothing, and I will tell you that to your face rather than sell you a detox protocol on the strength of a deletion that half the population carries.

Tryptase, plasma histamine, and 24 hour urine N-methylhistamine, if mast cell features dominate

What it measures. Mediators released by mast cells, measured in blood and in a 24 hour urine collection.

Why I order it. Only when the picture is mast cell shaped, with flushing, hives, abdominal cramping, blood pressure swings, and multisystem reactions arriving in episodes rather than continuously.

How to read it. The timing of the draw matters more than the number. The review underlying the modern criteria describes a substantial rise in serum tryptase during an attack compared against the person's own baseline tryptase, which is an acute-versus-baseline design, not a single-value threshold (PMID 34166845). A tryptase drawn on a good day is answering a different question than the one you asked. Plasma histamine and 24 hour urine N-methylhistamine I read as adjuncts, and I will say plainly that I do not have the same quality of support for them as for tryptase.

I publish no functional targets for these three. The laboratory's reference range and the interpretation of the clinician managing that part of your care govern, and a genuine mast cell disorder is a specialist evaluation, not something a functional panel settles.

What changes. A meaningful rise during a flare over your own baseline sends you to a specialist evaluation, and that referral outranks everything else in this plan. It also raises the priority of the mast cell arm of the protocol, which in my hands is quercetin, vitamin D, and omega-3. On quercetin I owe you a specific honesty: the closest human trial at exactly the doses I use, 500 mg and 1000 mg a day for 12 weeks in 120 women, raised blood quercetin substantially and produced no measurable change in innate immune function or in the inflammatory markers it measured (PMID 20500927). It did not measure mast cell mediators, so it does not directly disprove mast cell stabilization. It is still the dose-matched trial, and it found nothing. I still use it, because it is low risk and some patients clearly do better on it, and you should know that is clinical experience rather than trial evidence.

Part 3: The numbers

Functional targets I publish

These are my clinical targets, used at Root Health. They are narrower than the standard laboratory range on purpose, and they are not validated diagnostic thresholds. Where the standard column says your laboratory governs, it means exactly that.

Marker	Units	Functional target	Approximate standard range
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Ferritin	ng/mL	50 to 150	25 to 300
Homocysteine	umol/L	5 to 7	under 10 to 15, varies by lab
hs-CRP	mg/L	under 0.5	under 1.0
TSH	mIU/L	1.0 to 2.5	your laboratory's range governs
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
Fasting insulin	uIU/mL	2.6 to 5.0	your laboratory's range governs

Vitamin B12 and red blood cell folate appear here because they are the first two inputs I check when homocysteine is high, not because they are separate baseline tests. Glucose, hemoglobin A1c, and fasting insulin are the follow-up when the metabolic panel or your symptom pattern points at fuel stability.

One number sits outside this table because it is a standard laboratory reference range rather than a functional target of mine. **Whole blood vitamin B1 runs 70 to 180 nmol/L of thiamine diphosphate, and a result below 70 suggests deficiency.** A result inside that range does not on its own rule out a functional shortfall, which is why I read it next to your symptoms rather than alone.

Where I publish no target, and say so

For the following I have no functional target that I can defend across every program on this shelf, so I print none rather than invent one. Your laboratory's reference range and your clinician's interpretation govern each of them.

RBC magnesium. Serum zinc, serum copper, and ceruloplasmin. Serum iron, total iron binding capacity, and transferrin saturation. Free T4 and free T3. AM cortisol and the four-point salivary curve. Prolactin. Whole blood or RBC glutathione and the GSH to GSSG ratio. Plasma and RBC amino acids. Urinary organic acids, including pyroglutamate. Urinary mycotoxins. Blood and unprovoked urine metals. Tryptase, plasma histamine, and 24 hour urine N-methylhistamine.

That list is longer than the one above it, and that is the honest shape of this field. A guide that printed a confident target for every one of those numbers would be telling you something nobody knows.

Part 4: Reading your results as a pattern

Individual markers mislead. Patterns are what I actually act on.

Pattern one: an ordinary treatable cause found. Iron deficiency, a thyroid abnormality, a genuine vitamin D deficiency, a functional B12 deficiency caught by methylmalonic acid. This is the best outcome this workup can produce. It gets corrected with your physician, on its own terms, before anything else starts.

Pattern two: the glutathione system depleted, cofactors short. A low glutathione with a poor GSH to GSSG ratio, a whole blood B1 below 70 nmol/L, low glycine or cysteine on the amino acids. This is the pattern the protocol was built for, and it is also the pattern that most tempts everyone involved to move too fast. The sequence does not change. Stabilize and open drainage, then support the pathways, then and only then consider mobilizing and binding. One agent every three to five days, starting at a quarter to half the target dose, hypoallergenic formulations without fillers or dyes.

Pattern three: everything normal, symptoms entirely real. Common, and not a dead end. It moves the work to the parts of this condition that no blood test measures, which are exposure reduction, sleep, pacing, nervous system regulation, and food. Those are not the consolation prize. Reducing your exposures is the foundation of the whole plan and it never depended on a lab result.

Pattern four: mast cell features dominating. Episodic multisystem reactions with flushing and blood pressure swings deserve a specialist evaluation regardless of what the rest of the panel shows, and that referral takes priority over any supplement in my protocol.

Pattern five: mycotoxin or metal findings with a matching exposure history. The building comes before the binder, every time. Moisture correction, source removal, and air handling are the interventions with the

actual leverage, and no urine panel changes that ordering.

Part 5: The conversation with your doctor

A request list you can bring

Baseline

- Complete blood count with differential
- Comprehensive metabolic panel
- RBC magnesium
- Serum zinc and serum copper with ceruloplasmin
- 25-OH vitamin D
- Ferritin with a full iron panel
- Homocysteine, read as a B12, folate, B6, thyroid, and kidney workup trigger
- hs-CRP
- TSH, free T4, free T3
- AM cortisol
- Prolactin

Functional, where the history supports it

- Urinary organic acids, with methylmalonic acid and pyroglutamate specifically
- Plasma or RBC amino acids
- Whole blood or RBC glutathione with the GSH to GSSG ratio
- Vitamin B1, whole blood (LabCorp test 121186), for thiamine status, collected as whole blood and not plasma
- Urinary mycotoxin panel, understanding the guideline position stated above
- Heavy metals by blood, unprovoked, and only where exposure history supports it
- GSTM1, MTHFR, COMT, and SULT as supportive context, not as a diagnosis
- Tryptase, plasma histamine, and 24 hour urine N-methylhistamine if mast cell features dominate, drawn during a flare and compared against your own baseline

The question to ask before any panel is ordered

Ask the ordering clinician to state, before the sample is collected, what each possible result would mean and how it would change your care. If that question cannot be answered in advance, the test is not ready to be ordered. Hold me to it too.

Working with Root Health

If you want help reading your results and building the sequence around them, that is exactly the conversation the Root Discovery Session is for. It is free.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, start with two weeks of honest tracking. Symptoms, exposures, sleep, food, and function, dated. That log is the endpoint every number in this guide gets held to, and it costs nothing.

Free tracker: rootcauseunlocked.com/tracker

I keep a screened dispensary through Fullscript, which ships directly to you:
us.fullscript.com/welcome/drseay. Root Health earns a portion of dispensary purchases.

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Every claim above that rests on a specific study carries its PubMed ID inline. You can read any of them by entering the number at pubmed.ncbi.nlm.nih.gov. Listed here by identifier, design, and year, in the order they appear.

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Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

Educational content only. Not medical advice. Not a diagnosis. Never stop, skip, or change a prescription based on this guide. If a reaction is escalating or you are having trouble breathing, seek emergency care now.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose Multiple Chemical Sensitivity or anything else, and reading it does not make you my patient.

Symptoms you attribute to a chemical exposure can also come from asthma, migraine, heart rhythm disorders, carbon monoxide, medication reactions, thyroid and other endocrine disease, anemia, infection, panic, vestibular disorders, or occupational poisoning. New, severe, rapidly worsening, or unexplained symptoms get evaluated promptly, not supplemented. Ruling those out is the foundation this entire guide sits on, not a delay tactic.

Nothing here is a reason to stop, skip, or change a prescription. Every entry below is written as a possible addition to care you already have, coordinated with whoever manages it.

Pregnancy, breastfeeding, kidney disease, liver disease, cardiovascular disease, diabetes, seizure disorders, and multiple medications all change the safety math on this list. So does a history of reacting to supplements, which in this population is common.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

What you are buying, stated plainly

Not one supplement in this guide has been tested in a randomized controlled trial in people with multiple chemical sensitivity. Not one. Nobody has run those trials, and for compounds nobody can patent, nobody is likely to fund them.

That is a real gap and I am not going to paper over it. Here is what does exist, and I label every agent below with which of these it has:

- **Tested here.** A randomized trial in people with chemical sensitivity. Nothing in this guide reaches this level.
- **Tested elsewhere.** Randomized human trials, in a different condition or population.
- **Human biochemistry.** The pathway is established in living humans, and no outcome trial has been run on it.
- **Animal or test tube only.** Real mechanism, no human data.
- **Nothing found.** Searched, and the literature is empty or points the other way.

What that leaves is reasoned clinical practice held to an endpoint. I pick agents whose pathway is measurably relevant to you, start them one at a time, define in advance what should change and by when, and stop the ones that do not earn their place. That is a defensible way to work. It is not the same thing as proof, and I will not present it as proof.

The cardinal rule: start low and go slow, one agent at a time

People with chemical sensitivity react to supplements. That is not a caricature, and it is the single most important operating instruction in this guide.

The paper everyone cites for it deserves to be quoted accurately, because the accurate version is more useful than the loose one. In an Italian retrospective case series of 41 patients with multiple chemical sensitivity, 7 patients received **intravenous** reduced glutathione by continuous infusion, and 5 of those 7 had adverse reactions (PMID 24367721). That is a 70 percent reaction rate to an infusion, not to an oral supplement. It does not tell you that a capsule of glutathione will hurt you. It does tell you that in this population, the fastest, highest-dose, least reversible route into the body is where trouble shows up, which is exactly the argument for the slow oral approach in this guide rather than an argument against supplementing at all.

So the rule, with no exceptions I will endorse:

1. **One new agent at a time.** Never two. If two go in together and something happens, you have learned nothing and lost both.
2. **Start at a quarter to a half of the target dose,** and only move up once you have been steady for several days.
3. **Three to five days minimum before adding anything else.** Longer if you are in a flare, traveling, sleeping badly, or recovering from an exposure.
4. **Write down what you took, when, at what dose, and what happened.** Memory is not reliable in a symptom pattern this variable, and the log is what makes the next decision better than a guess.
5. **A supplement that cannot name what it is supposed to change, and by when, does not get started.**

Two gates sit in front of rule 1, and both are new.

Gate one, the workup before the supplements. The iron, B12 and bleeding-source workup happens BEFORE this sequence starts, not alongside it. If iron is low, we find out why it is low before we replace it, because replacing iron without finding the loss means replacing it again in six months. Nothing in section one begins until that workup is done.

Gate two, two weeks of a stable diet before the first agent. Not one week. This is the gate that makes everything downstream readable, and the reason is specific to this condition. Across thirty-seven provocation studies covering 784 people with chemical sensitivity, reactions tracked whether the person could tell the real exposure from the sham, and masked or below-threshold exposures did not produce consistent responses (PMID 17137865, reproduced in PMID 18568800 and PMID 8234918). In a condition where attribution to a specific agent is that unreliable, introducing a new supplement every three to five days on top of a moving diet manufactures false conclusions in both directions. Stabilize the input first. Then the log means something.

When you feel worse

Some people feel worse when they start something new, and the wellness world has a ready explanation: worsening proves the protocol is working. I do not accept that framing. A framework in which feeling worse is always evidence of progress cannot be tested and cannot be wrong, which makes it useless for deciding anything.

Worsening is information. Sometimes it settles within a few days. Other times the dose is too high, or the groundwork was skipped and the agent arrived out of order. Sometimes the product is simply not right for you, and sometimes it is an excipient rather than the active ingredient. Significant worsening is a reason to stop and reassess with me, not a milestone to push through.

1. The glutathione system

This is the centerpiece of the protocol, and it has the most direct evidence that something is actually different in this population. Two independent research groups have measured the glutathione system in people meeting clinical criteria for multiple chemical sensitivity and found it depleted (PMID 40943945, PMID 20430047). The nutrients that system runs on are established human biochemistry, not speculation.

The honest limit, stated once and applying to everything in this section: no trial has tested whether replacing those nutrients makes anyone with chemical sensitivity feel better, and a low glutathione level can be a consequence as easily as a cause. So I treat repletion as a measurable, mechanistically reasoned starting point that we recheck, not as a proven therapy.

N-acetylcysteine, 600 mg once daily, titrating to 600 mg two or three times daily as tolerated

What it does. Glutathione is a three-part molecule built from glutamate, cysteine, and glycine, and the enzyme that assembles it is the rate-limiting step (PMID 18601945). Cysteine is the part that runs short first, and NAC is a cysteine donor. Of everything in this section, this is the most direct way to hand your body the raw material.

Why it is here. It is the first agent I start, because it addresses the specific system that was measured as depleted, it is inexpensive, and it is the one with the most human data behind the general effect.

Evidence: tested elsewhere. A pooled analysis found glutathione levels rose after NAC supplementation, along with lower markers of oxidative damage (PMID 39632267). The trials in that pool were in exercise, lactate, and muscle-damage research in largely healthy or athletic people, not in chemically sensitive patients. So the effect on glutathione is genuine and measured in humans, and the population is not yours.

How to introduce it. Start at 600 mg once daily, in the morning, with food. Hold there for at least a week before considering a second dose. NAC commonly causes gastrointestinal upset, it interacts with some medications, and it can rarely provoke bronchospasm, which matters more than usual if you have reactive airways. If it does not sit well, that is worth knowing on day three rather than after you have layered three other things on top of it.

Liposomal or acetyl-glutathione, 500 to 1,000 mg per day

Why this number changed. My earlier protocol used 100 to 250 mg per day. Nobody has studied that. The trial that exists randomized 54 healthy adults to plain oral glutathione at 250 mg per day or 1,000 mg per day for six months and found whole blood glutathione rose 17 percent and red cell glutathione rose 29 percent (PMID 24791752). A dose nobody has tested is a guess, not a clinical position, so I moved the number to where the evidence actually is.

What it does. It supplies glutathione directly rather than supplying the parts and hoping assembly keeps up.

Evidence: tested elsewhere for plain oral glutathione, animal or test tube only for the acetylated form. Everything hinges on this distinction. The six-month randomized trial above used plain oral reduced glutathione. For S-acetylglutathione specifically there is neither a human trial nor any human pharmacokinetic data, only cultured human fibroblasts and a mouse model (PMID 15617191, PMID 14624358). Any claim that the acetylated version outperforms plain glutathione is currently unsupported in people. **If your goal is raising measured glutathione at this dose, plain oral glutathione is the form with the human trial behind it,** and it is usually the cheaper one.

How to introduce it. This is the second agent, not the first, and it goes in after NAC has been steady. Start at a quarter to a half of target. Take it away from food if the label says to.

A caution on the inhaled and nasal routes. You will see nebulized and intranasal glutathione promoted for exactly this population, and the delivery genuinely works: nebulized glutathione raises glutathione in the airway lining several fold (PMID 14726422), and intranasal glutathione measurably raises brain glutathione on imaging (PMID 28725693). The problem is precisely the population in question. In a randomized crossover trial in patients with mild asthma, nebulized glutathione caused a 19 percent drop in FEV1 with cough and breathlessness, attributed to sulfite formation (PMID 9279219). **Reactive airways are a reason to avoid the nebulized route, not a reason to choose it.** The larger trials in cystic fibrosis and Parkinson's delivered the drug and did not produce clinical benefit (PMID 23631796, PMID 28436395). I do not use the nebulized route in this protocol.

Selenium, 100 to 200 mcg per day as selenomethionine

What it does. Selenium is the required trace element for glutathione peroxidase, the enzyme family that actually spends glutathione to neutralize peroxides.

Why selenomethionine. There are direct human head-to-head comparisons of selenium forms, and they support this form with one important qualifier. Selenomethionine is incorporated into body protein wherever methionine would go, so it raises measured blood selenium and builds a reserve in a way inorganic selenite does not (PMID 8909692, PMID 2484510).

Evidence: tested elsewhere, and the qualifier matters. Selenomethionine is not better at driving the actual enzymes. In selenium-replete adults given roughly 200 mcg per day for 16 weeks, neither glutathione peroxidase activity nor selenoprotein P responded to supplementation at all, in any form (PMID 16614127), and a four-arm human study found all forms reached the same glutathione peroxidase steady state within a month (PMID 2484510). **The practical reading: selenium is worth taking to correct and monitor a genuinely low level. If your level is already normal, this dose raises a number on a report without moving the enzyme.** That is why selenium in my protocol is status-driven rather than automatic.

How to introduce it. Once daily with food, at the low end of the band, after the glutathione agents are settled.

Alpha-lipoic acid, 100 mg per day titrating to 300 to 600 mg per day

What it does. Human red blood cells take lipoic acid up and reduce it to dihydrolipoic acid in a glucose-dependent, NADPH-dependent reaction catalyzed partly by glutathione reductase. We know that step happens in humans because red cells from a person with G6PD deficiency could not perform it at all (PMID 7632170). Dihydrolipoic acid is then chemically capable of regenerating other antioxidants.

Evidence: human biochemistry, with randomized downstream results. The uptake and reduction step is established in human tissue. Randomized trials at 200 to 600 mg per day have shown downstream markers of oxidative damage improve (PMID 19231043, PMID 22374556). What has never been directly measured inside a living person is the recycling step itself, so the antioxidant network story is a well-grounded inference rather than a demonstrated result. Oral pharmacokinetics show brief peaks and roughly 30 percent bioavailability (PMID 14551180), which is why it is dosed on a schedule rather than once and forgotten.

On the mercury question, which is usually described wrong. You may have been told lipoic acid mobilizes mercury and redistributes it into the brain. Here is what was actually shown. In rats, intravenous lipoic acid increased biliary excretion of **inorganic** mercury 12-fold to 37-fold, and in the same study it **decreased** biliary excretion of methylmercury, cadmium, zinc, and copper (PMID 1585376). The movement demonstrated was into bile for elimination, which is the direction you want. Redistribution into brain was not demonstrated. **Evidence: animal only**, at intravenous doses far higher per kilogram than an oral capsule, and no human study has measured any of it. I am not going to inflate an animal finding into a warning it does not support, and I am also not going to pretend an unstudied question is settled. What I do is keep the titration slow and keep binders and mobilization out of the early phase, which is the same sequencing I would use anyway.

How to introduce it. Start at 100 mg per day. It can lower blood glucose, so if you are on medication for diabetes this is a conversation with your prescriber before the first dose, not after.

Milk thistle, silymarin 150 to 300 mg standardized, twice daily

What I do not claim. Milk thistle is sold as a phase I and phase II balancer. That claim comes from the test tube, where silymarin lowers CYP3A4 and glucuronidation activity in human liver cell cultures (PMID 11038151, PMID 33706132). When it was tested in actual people using validated probe drugs, including at 420 mg of silymarin per day for two weeks, it produced no measurable change in CYP1A2, CYP2C9, CYP2D6, CYP2E1, or CYP3A4 (PMID 25028567, PMID 15536458, PMID 18214849). **Evidence for the detox-pathway claim: animal or test tube only, and contradicted in humans.** I do not tell patients it balances their detoxification pathways.

Why it is still in this section. It sits here for the glutathione endpoint, not the enzyme one, and that endpoint has randomized human support: silymarin raised erythrocyte glutathione in patients with alcoholic liver disease (PMID 11841050) and raised plasma glutathione substantially against placebo in a second, independent randomized trial (PMID 29154063). **Evidence for the glutathione effect: tested elsewhere.** Two honest mismatches come with it. Both trials dosed three times daily, at 420 to 450 mg per day, while my protocol gives 150 to 300 mg twice daily. And both populations had liver disease, which is not you. The reassuring flip side of the negative probe-drug studies is genuine: measurable silymarin exposure with no measurable CYP shift is the best available answer on drug interaction risk.

How to introduce it. This is the last agent I add in section one, and the first one I drop if we are simplifying. Start at 150 mg once daily with food.

2. Phase I and phase II conjugation cofactors

Detoxification is not one process. Phase I chemically activates a compound, phase II attaches something to it so it can leave, and the enzymes doing both run on specific vitamins and minerals. When those run short, the machinery does not stop cleanly, it runs badly.

Thiamine, and why it sits upstream of everything above

Start here, not with glutathione. This is the piece most protocols put in the wrong place. Regenerating used glutathione requires NADPH. A major source of NADPH is the pentose phosphate pathway, and the rate-limiting enzyme of its non-oxidative branch is transketolase, which cannot function without thiamine (PMID 34685573). If thiamine is short, the NADPH supply is short, and every glutathione agent in section one is being poured into a system that cannot recycle what it already has.

Evidence: human biochemistry, no outcome trial in this population. The transketolase dependency and the NADPH link are established. There is no clinical trial of thiamine in multiple chemical sensitivity, and I will not imply otherwise.

The test matters more than usual here, and so does ordering the right one. Serum thiamine is a poor read on functional status. The test I order is **LabCorp test 121186, Vitamin B1 Whole Blood**, the Mayo equivalent of which reports thiamine diphosphate, the active coenzyme form. The reference range is **70 to 180 nmol/L of thiamine diphosphate, and a result below 70 suggests deficiency** (PMID 34685573). Two details are worth saying plainly. It has to be drawn as **whole blood, not plasma**, because the coenzyme sits inside the red cells and a plasma tube throws away most of what you are trying to measure. And a normal blood level does not exclude a functional deficiency. That second limit is the exact problem the older literature was chasing when it named the erythrocyte transketolase activation test, which measured how much the enzyme sped up when thiamine was added to the sample rather than how much thiamine was sitting there. You will see that assay named in the papers, and you should know it is not commercially orderable in the United States, so it is not something your doctor can put on a requisition.

On the dose. The form I use is **benfotiamine, 150 mg twice daily, for 4 to 6 weeks**, and then we look again rather than continuing indefinitely on a theory. Because a normal level does not close the question, I judge the trial on your symptoms and your response alongside a repeat level, not on the number by itself. **Evidence: established biochemistry, no outcome trial in this population.** That is the same tier as the paragraph above, and I will not quietly upgrade it because a specific dose is now printed.

Magnesium glycinate, 200 to 400 mg elemental per day

What it does. Magnesium is a cofactor for hundreds of enzymatic reactions, including steps in energy production that everything else in this protocol depends on. The glycinate form is chosen for tolerability and because glycine is itself useful here (see below).

Evidence: nothing found in this population. There is no trial of magnesium in chemical sensitivity, and I will not invent one. What I do instead is straightforward: I check red cell magnesium rather than serum, correct a genuine shortfall, and judge it on the things magnesium actually and reliably improves, which are bowel regularity, muscle symptoms, sleep, and migraine if you have it. If none of those change, it has not earned a permanent place.

How to introduce it. Evening dosing, starting at the low end. Too much magnesium causes loose stools, which is the built-in dose signal. It is unsafe in significant kidney failure.

Vitamin C, 250 to 500 mg per day total

Why this number moved down. My protocol said 500 mg two or three times daily. The pharmacokinetics do not support it. NIH depletion and repletion work shows that single doses at or above 500 mg have declining bioavailability with the surplus excreted, immune cells saturate at about 100 mg per day, plasma saturates at 1,000 mg per day, and oxalate and urate excretion rise at that same 1,000 mg (PMID 8623000, PMID 15068981). A higher split-dose regimen was buying expensive urine and a stone risk, so the number came down.

Evidence: tested elsewhere, at a specific dose. This is one of the few places where somebody has actually tested supplementing against a defined environmental oxidant exposure. Randomized trials in ozone-challenged healthy adults and in ozone-exposed asthmatic children in Mexico City used **250 mg of vitamin C per day together with vitamin E** and found benefit on lung function measures (PMID 11549539, PMID 12204869, PMID 15498043), while a separate trial in ozone-responsive subjects found no protection against symptoms (PMID 16767844). So the exposure-dosing concept survives, the benefit favored function over symptoms, and the honest note is that every positive trial paired vitamin C with vitamin E rather than using it alone. Vitamin E is not in this protocol; if you want that combination, it is worth raising with me rather than adding on your own.

Vitamin E now goes with it, which closes the gap named above. Every positive trial at this node paired vitamin C with vitamin E rather than using either alone, so the pair is what I use. Systematic review evidence supports co-administration of the two on plasma oxidative markers (PMID 40740780), with randomized evidence for the combination (PMID 34122682). Vitamin E alone is not the intervention that was tested and is not what I recommend.

One dose note, because it differs by condition and the difference is real. The 250 to 500 mg figure here is the chemical-sensitivity dose, and it comes from the ozone-exposure trials above, which is the closest exposure model to this condition. It is deliberately not the same as the 2 g per day used where histamine clearance is the target. Those are two different questions with two different literatures, and carrying one dose across to the other condition would be exactly the kind of error this guide exists to avoid.

How to introduce it. A single daily dose, with food, alongside vitamin E. If you want a pre-exposure dose ahead of something you know will be difficult, that is reasonable within the same daily total, not on top of it.

The active B vitamins, from riboflavin and P5P to methylfolate and B12

Riboflavin is the one with the real mechanism, and the dose honesty to go with it. Riboflavin becomes FAD, and FAD is the required cofactor for glutathione reductase, the enzyme that converts used glutathione back into the working form. That is why riboflavin status is measured functionally through that very enzyme (PMID 12791609, PMID 19102813, PMID 37603429). FAD and FMN are also the two flavin cofactors of NADPH-cytochrome P450 oxidoreductase, the electron donor for every microsomal phase I reaction (PMID 22982532). So riboflavin status is rate-relevant to both glutathione recycling and phase I metabolism, in humans, on established biochemistry.

Evidence: human biochemistry, with a dose caveat you should know about. The genotype-targeted human trials used **1.6 mg per day** (PMID 23608654, PMID 22277556). The 25 to 50 mg found in practitioner B-complexes, including the one I use, is a professional convention rather than a studied amount, and no human data show added cofactor benefit above the low single-digit milligram range. I keep the higher amount because that is how active B-complex products are actually formulated and because riboflavin has a wide safety margin, and you should know that the number is convention, not trial evidence.

B12, and the three things that go with it. B12 goes in at **2,000 mcg per day**, and it does not go in alone. Folate, B6 in its P5P form, and TMG (trimethylglycine) go alongside it. The reason is the shape of the pathway rather than a preference: homocysteine is cleared by more than one route, B12 and folate drive one of them, TMG drives the other, and B6 governs the exit into transsulfuration, so supplying one input and not the others is how people end up stalled with a number that will not move. **Evidence: established human biochemistry for the pathway, no outcome trial in this population.** The B12 number is the one I hold to. The folate, B6, and TMG amounts follow the product and your labs, and I set those with you rather than printing one figure here for everyone.

On MTHFR, because you will be asked about it. This one is genuinely established. The C677T variant destabilizes FAD binding, and randomized genotype-targeted trials show that adults with the 677TT genotype respond to riboflavin when other genotypes do not, with a measurable shift in DNA methylation specific to that genotype (PMID 23608654, PMID 32334045). Genotype really can change how a person responds to a B vitamin. That is a proven human fact, and it is one of the few genetic claims in this space that is.

On COMT, the honest other half. The COMT Val158Met enzyme activity difference is well established in human brain tissue, and the enzyme does consume methyl groups (PMID 8807664, PMID 15457404). What no study has tested is whether COMT genotype predicts who reacts badly to methyl donor supplements. **Evidence: human biochemistry for the enzyme, nothing found for the prediction.** I use it as a hypothesis that makes me start lower, never as an explanation I hand you as fact.

The overmethylator question. If you flare on methylfolate or methylcobalamin, I switch you to hydroxocobalamin and folinic acid. You deserve to know exactly what that decision is made of. **Evidence: nothing found.** Nobody has published a definition of an overmethylator or a validated test that identifies one, and no study has compared hydroxocobalamin against methylcobalamin in people who react badly to B vitamins (PMID 27068282). Folinic acid is a real and well-studied folate form, but the condition it is proven for is a specific antibody-mediated folate transport problem, which is not the same thing as reacting to methyl donors (PMID 27752075, PMID 29661558). One more point, and it has two halves. My functional target for homocysteine is 5 to 7 umol/L, and I want you to hear that lower is not automatically better: a homocysteine driven down below that band is the point where I start watching for overmethylation and B vitamin intolerance, and I would rather slow the

methyl donors than keep pushing a number down. That is my clinical position and I am labeling it as one, because the evidence half is thin. No published reference interval or validation study assigns an overmethylation meaning to a low homocysteine, and a low value most often reflects good B vitamin status or brisk clearance toward cysteine (PMID 20301697). So a low number by itself does not make me change your B vitamins. It makes me watch, and how you actually feel on them decides the rest.

So the switch is clinical convention, and I make it anyway, for a reason I will state plainly: the non-methylated forms are equally capable of correcting a deficiency, the switch costs nothing, and if you consistently feel worse on one form and fine on the other, that observation is worth more to your plan than a label. I am acting on your response, not on a provable phenotype.

Glycine, 3 g per day

What it does. Glycine is one of the three amino acids glutathione is built from, and it is the one that becomes rate-limiting in some populations. It is also the conjugating partner for a whole class of phase II reactions.

Evidence: tested elsewhere, at a higher dose and never alone. The trial usually cited for glycine and glutathione gave roughly 100 mg per kilogram per day, which is about 7 g per day for a 70 kg adult, and it gave glycine together with cysteine rather than by itself (PMID 21795440). My 3 g is under half the studied amount, and it was chosen for tolerability in a population that does not do well with large single ingredients. You should know the number is a practical compromise rather than the studied dose.

How to introduce it. Powder is usually easier than capsules at this amount. Evening dosing suits most people.

Taurine, 500 to 1,000 mg twice daily

What it does. Taurine is a sulfur-containing amino acid and the conjugating partner for bile acids, which is one of the routes by which conjugated compounds actually leave the body rather than circulating.

Evidence: nothing found in this population, and thin elsewhere. The human evidence I can point to for the bile conjugation half comes from a small study in children with cystic fibrosis and severe fat malabsorption, which is about as far from your situation as a population gets. Nothing establishes a taurine benefit in chemical sensitivity. I keep it because the bile route is the exit path for a large share of phase II output and because taurine is well tolerated at this dose, and I judge it by whether the gut and the post-exposure recovery window improve, not by a marker.

Zinc, 15 to 30 mg per day, with 1 to 2 mg copper

What it does. Zinc is a cofactor for a long list of enzymes, including alcohol dehydrogenase and the metallothionein system, and it matters for immune regulation and barrier repair.

Evidence: nothing found in this population. I want to correct something you may have read, including from me. Zinc deficiency is often called one of the most consistent findings in chemical sensitivity. The largest direct mineral study, a case-control study of 408 women meeting validated inclusion criteria, found **no statistically significant differences** in intra-erythrocytic minerals, and zinc was one of six minerals that trended lower without reaching significance or showing a dose response (PMID 17046989). That is not a zinc-specific finding, and I no longer describe it as one. Zinc goes in when your level says it should, not because of the diagnosis.

The copper is not optional. Zinc supplementation depletes copper. In the classic human study, red cell copper-zinc superoxide dismutase fell significantly after 10 weeks of zinc, and the authors stated plainly that zinc supplementation poses a risk to iron and copper status (PMID 2912000). That enzyme is an antioxidant enzyme, which means unaccompanied zinc can quietly work against the entire point of this protocol. If zinc goes in at this dose, copper goes with it.

3. Neurogenic and sensory reactivity

This section used to be headed “mast cell and histamine support,” and I changed it because the evidence does not put histamine in the driver’s seat here.

Only one study has measured histamine in people with self-reported chemical sensitivity. It found substance P, vasoactive intestinal peptide and nerve growth factor elevated in those patients, **and histamine not elevated**. Solvent exposure then raised all of them in the patients but not in controls, and made the skin more reactive to histamine applied directly (PMID 15031958). Those are nerve signaling molecules. Substance P activates mast cells, so there is a genuine route from an irritated sensory nerve to a degranulating mast cell, but it runs nerve first and mast cell second. Histamine is a passenger here, and sometimes an amplifier. It is not the engine.

So the agents below stay, because several of them have work to do, and the reason they are here has changed. They are aimed at the reactivity of an irritated sensory system, not at blocking a histamine flood that the measurements do not show.

Two groups of people should still read it. If your reactions arrive within minutes with flushing, hives, sudden gut symptoms and a racing heart, that pattern deserves its own workup rather than being managed as a corner of this one, and I would rather send you down that road properly. For everybody else, this is a supporting section rather than the main event, and the leverage lies in section 1.

Quercetin, 200 mg per day of a bioavailable form

The dose and the form both changed, and the form is the part that matters. My protocol said 500 mg twice daily of unspecified quercetin. The two positive randomized human trials used **bioavailable preparations at 200 mg per day** (PMID 19295240, PMID 35776034). Plain quercetin aglycone absorbs poorly enough that a larger dose of it can deliver less active compound than a smaller dose of a good form. Read the label for the form, not the milligram number.

What it does, and why it belongs in this section specifically. Quercetin attenuates mast cell degranulation through **MRGPRX2**, binding the regulatory receptor CLM-1 with affinity measured directly by surface plasmon resonance (PMID 39398230). That matters here more than anywhere else on the shelf, because MRGPRX2 is the receptor that responds to compounds with no allergy involved at all, which is the mechanism behind reactions that test negative for allergy and get dismissed. It also negatively regulates the allergic route through CD300f, reducing histamine and inflammatory mediators in a urticaria model (PMID 40309955), against a broad anti-allergic literature (PMID 27187333). The older in vitro comparison against cromolyn stands (PMID 22470478).

The honest limit, which has not changed. A 12-week human trial at 500 and 1,000 mg per day raised plasma quercetin substantially and found no effect on innate immune function or on IL-6 and TNF-alpha (PMID 20500927). That trial used the old dose and did not measure any mast cell mediator, so it is evidence about immune markers rather than evidence against the mechanism above. And quercetin has no evidence for raising diamine oxidase, so it is never presented as a histamine-clearance agent.

So it still earns a defined trial with a defined endpoint: reaction frequency and severity in your log over four to six weeks, at the corrected form and dose.

How to introduce it, and one hard rule. Half dose to start, with food. **Never in the same dose as iron, and never with an iron-rich meal.** Quercetin is a polyphenol and it inhibits iron absorption, which matters enormously in a population where low iron is common. Separate them by at least two hours. It also interacts with several medications, so it belongs on your medication review list.

Vitamin D

Both positions, because you are entitled to both. The 2024 Endocrine Society guideline suggests against routine 25-OH vitamin D screening in healthy adults and states that no clear evidence defines an optimal level (PMID 38828931). The trials support that caution: pushing healthy adults into the 40 to 60 band produced no added benefit and slightly lower radial bone density (PMID 31454046), and the large VITAL trial at 2,000 IU daily was null on its primary outcomes (PMID 30415629).

And here is mine. I test 25-OH vitamin D and I use a functional target of 40 to 80 ng/mL. The distinction worth drawing is that a guideline about screening asymptomatic populations is not the same question as case-finding in a symptomatic person who is already in my office. Where I agree with the guideline entirely is on chasing a number: I treat a genuinely low level, and I do not push someone who is already sufficient higher because a chart says so. **Evidence: tested here in the sense that the trials exist and are largely negative for supplementing the already-sufficient.** Dose follows your result, which is why there is no number printed in this paragraph.

Omega-3 fats, filed for what they actually do

A note on placement first. Omega-3 sits in this section for convenience, not because it is a mast cell agent. The one human trial that measured mast cell mediators directly, at 4 g EPA plus 2 g DHA daily for three weeks, changed neither bronchial reactivity nor urinary mast cell mediator excretion despite confirmed adherence (PMID 25321659). What omega-3 does have is the systemic inflammatory evidence below. I would rather name the benefit accurately than let a heading imply one it does not support.

What they do. The mechanism is established at several independent levels in humans: membrane phospholipid remodeling, substrate competition with arachidonic acid, and the role of EPA and DHA as the obligate raw material for the resolvins and protectins that actively end an inflammatory response rather than just suppressing it (PMID 25149823).

Evidence: tested elsewhere, with the dose corrected downward. An umbrella analysis of 32 meta-analyses shows significant reductions in CRP, TNF-alpha, and IL-6 across varied adult populations (PMID 35914448). The dose is where the older protocol was wrong. A dose-response meta-analysis of 40 randomized trials found the CRP benefit is non-linear and plateaus near **1,200 mg per day of combined EPA and DHA**, with no significant effect in overweight and obese participants (PMID 40263171). My protocol previously said 2 to 3 g combined, which sits above where the dose-response evidence lands, so I start near 1.2 g of combined EPA and DHA and require a specific reason before going higher.

How to introduce it. Read the label for the combined EPA and DHA number rather than the capsule size, which is usually much larger. At supplement doses omega-3 has a blood-thinning effect worth naming to your prescriber if you take anticoagulants or have surgery scheduled. Fish oil is also a common source of reflux and burping, and in this population a rancid or heavily flavored product is a frequent unnecessary reaction.

4. Mitochondrial support

CoQ10 as ubiquinol, 100 to 200 mg per day

What it does. CoQ10 is the obligate lipid-soluble electron carrier between complexes I and II and complex III of the respiratory chain. Human primary CoQ deficiency syndromes prove that pathway is rate limiting in people, not just in theory.

Evidence: tested elsewhere, and this is the strongest evidence in the guide. Both halves of what I use it for have meta-analyzed randomized evidence. Thirteen randomized trials in 1,126 participants show CoQ10 reduces fatigue scores (PMID 36091835). Six randomized trials in 371 adults show it cuts migraine frequency by roughly 1.5 attacks per month and shortens attack duration, **though not severity** (PMID 33402403).

The dose caveat, stated plainly. The anchoring positive migraine trial used 300 mg per day, given as 3 doses of 100 mg (PMID 15728298). The only migraine study at 100 mg was open-label rather than blinded (PMID 27670440). And the fatigue meta-regression found the benefit grew as the daily dose rose. So **100 to 200 mg per day is a reasonable starting dose rather than the dose that generated the evidence**, and if it is working partially at 200 mg, that is an argument to discuss going higher rather than to conclude it failed.

On ubiquinol specifically. Whether ubiquinol is genuinely better than ordinary ubiquinone inside a person is not settled. The study showing ubiquinol superiority was small, non-randomized, and authored from a CoQ10 company (PMID 27128225), while an independent randomized three-period crossover found the ubiquinol advantage was not statistically significant and showed that CoQ10 appears in plasma mostly as ubiquinol even when swallowed as ubiquinone (PMID 32188111). Small trials have used ubiquinol at 100 mg and 150 mg in fatigue-related illness (PMID 23394493, PMID 27125909). I use ubiquinol because it is what the fatigue trials used, not because a head-to-head trial on a symptom outcome has ever been run. It has not.

How to introduce it. With a fat-containing meal. Morning dosing, because in some people it is mildly energizing.

Riboflavin

Covered in section two. It belongs to both sections, because the same FAD cofactor that regenerates glutathione is also a flavin the mitochondrial machinery depends on. It does not get counted or dosed twice.

Acetyl-L-carnitine, 500 mg twice daily, added last

What it does. Carnitine moves long-chain fatty acids into the mitochondria to be burned, and the acetylated form crosses into the nervous system more readily.

Evidence: nothing found in this population. There are randomized trials of acetyl-L-carnitine in other conditions. There is nothing in chemical sensitivity.

Why it goes last, and the honest source of that instruction. In my practice, acetyl-L-carnitine is occasionally activating, causing restlessness, a wired feeling, or disturbed sleep if it is taken late. I searched the literature for that effect specifically and could not find it documented. **So the activating caution is my clinical observation, labeled as clinical observation, not a documented finding.** It costs you nothing to act on it, so I act on it. Dose it early in the day, never in the evening, and add it after everything else in this protocol is stable so that if something changes, you know what changed it.

5. Gut and barrier support

An intestinal barrier that is doing its job determines how much of what you swallow, and of what your own microbes make, gets into circulation. This section comes late in the sequence on purpose, and one agent in it changed this year.

Colostrum, the primary gut agent

What replaced what, and why. L-glutamine used to hold this slot. It no longer does. Colostrum is the primary gut agent in this protocol as of 2026, because the trials in colostrum used 20 g per day for 14 days and **measured intestinal permeability directly**, which is the actual endpoint this section is about, rather than a marker one step removed from it.

Evidence: tested elsewhere. The permeability trials are not in chemically sensitive patients, and no trial has tested colostrum in this condition.

The introduction caution, which is not a footnote here. Colostrum is a dairy product, and it is going into a population where dairy reactivity, casein sensitivity, and general food reactivity are common. Two things follow. If you have a milk allergy, this agent is out, full stop, not started cautiously. If you have dairy intolerance rather than allergy, this is the agent in the entire protocol that most deserves the quarter-dose start, a full week at that dose, and an honest log. Do not start colostrum in the same week as anything else.

On glutamine, since I removed it. Glutamine's glutathione evidence is genuine, and I do not want it misremembered as absent. At least four randomized human trials used tissue or plasma glutathione itself as the endpoint and found glutamine raised or protected it, in surgical, trauma, and HIV-positive patients (PMID 12605586, PMID 19608826, PMID 22261571, PMID 18065204). The problem is route and dose. The strongest results used **intravenous** glutamine at roughly 0.5 g per kilogram per day, and the one oral trial used **20 g per day** (PMID 22261571). The 3 to 5 g in a typical gut capsule is a fraction of that, and it is not the amount anything was demonstrated at. If glutamine has a place for you, it is at the studied oral dose with a stated reason, not as a small background scoop.

Probiotics, where the strain matters more than the count

Read this part before you buy anything. Probiotics are the one item in this guide where a reasonable-looking product can make you worse, and the reason is specific rather than vague.

Some bacteria carry the gene for histidine decarboxylase, the enzyme that converts the amino acid histidine into histamine. A large genomic screen of 102,018 bacterial genomes found that about 3.6 percent carry that machinery, and when the authors measured actual output, gut-associated *Limosilactobacillus reuteri* sat in the highest producing group, alongside organisms nobody would put in a capsule on purpose (PMID 39029335). *L. reuteri* is sold as a probiotic. It is in products on open shelves right now.

Why this matters for you specifically. If histamine is part of your picture, and for many people in this population it is, then adding a histamine-producing organism to your gut is adding to the load you are trying to lower. Nothing about the product label will warn you.

The rule I follow. Histamine production is a **strain-level** trait, not a species-level one, so a label that reads only a genus and species with no strain identifier cannot be assessed at all. That single fact rules out most of the market. I do not use a proprietary multi-strain blend here, because a blend is a set of organisms you cannot individually evaluate, and if you react you will not know to what.

What I look for instead. A single named organism with its strain designation printed, so the question can actually be answered. On the other side of the ledger, strains that lack the histidine decarboxylase gene cannot make histamine, and some appear to degrade it: a *Lactiplantibacillus plantarum* strain characterized in 2026 degraded histamine by roughly 79 percent, and the genomic explanation was the absence of the *hdc* genes (PMID 42195967). Be clear about what that is: a laboratory characterization of one strain, not a trial in patients, and not a reason to expect a particular result in you.

Saccharomyces boulardii is the exception I am most comfortable with, and part of the reason is structural. It is a yeast, not a bacterium, so the bacterial histidine decarboxylase question does not apply to it at all. Its randomized evidence is genuine and worth knowing the shape of. It is strong for preventing antibiotic-associated and traveler's diarrhea, and it is listed as promising and needing more evidence for irritable bowel syndrome and other uses (PMID 20458757). None of it was done in chemical sensitivity.

Evidence tier, stated plainly. Nothing here has been tested in chemically sensitive patients. The histamine-production finding is genomic and laboratory work, and the degradation finding is a single-strain laboratory characterization. What I am doing is applying a mechanism that is well established to a population it has not been studied in, and choosing the option that cannot make the mechanism worse.

How I introduce it. Late, after the gut is otherwise settled, one product at a time, never alongside another new item. I judge it on what you can observe, meaning stool form, regularity, bloating, and how you feel after meals. I do not print a CFU number, because strain counts vary too widely between products for one figure to carry meaning, and I set it against the specific product you and I choose. If you react, we stop and we know exactly what we stopped, which is exactly why I use one named organism rather than a blend.

Binders, which are last and gated

Why a binder does anything at all, stated first, because it usually gets left out. Your liver conjugates a toxin and sends it out in bile. Bile empties into the gut. If nothing binds that toxin in the gut, a share of it is reabsorbed across the intestinal wall and returns to the liver, and the same molecule goes around again. That loop is called enterohepatic recirculation, and it is not a theory. Ochratoxin A, a mycotoxin, shows an oscillating rather than smoothly falling elimination curve exactly as a recirculating compound should, with 28 to 68 percent of the conjugated toxin recoverable in bile (PMID 3344528).

And a binder interrupts the loop. In that same work, a bile-binding agent abolished the oscillating pattern. In a controlled animal study, dietary cholestyramine lowered plasma ochratoxin A, reduced how much came out in bile and urine, **raised how much left in the stool**, and improved both kidney enzyme markers and kidney tissue appearance (PMID 9482354, with the same direction of effect in PMID 10606152 and PMID 8597155). The toxin left the body instead of being recycled.

In humans the same principle is demonstrable. Repeated doses of activated charcoal increase the clearance of substances already circulating in the bloodstream, through diffusion back across the gut wall onto the charcoal, with a dose-response up to a ceiling (PMID 7650764). Calcium montmorillonite clay has randomized human safety and efficacy data in aflatoxin-exposed children against a placebo (PMID 25135766, PMID 36133399), with earlier human safety evaluation (PMID 16019795, PMID 22249378). Clinoptilolite, the zeolite, has a randomized controlled trial in diarrhea-predominant irritable bowel syndrome (PMID 36569277).

So binders are not a folk remedy in this protocol. They act on a documented route, they are the last step for a reason, and the reason is that clearing bile only helps once the pathways that put things into bile are working.

What they do not do. They have no effect on histamine and none on diamine oxidase. If anyone sells you a binder as a histamine intervention, that is not what this evidence supports.

Two gates, both of which have to be open before a binder is started.

Gate one: bowel regularity. Binders go nowhere if transit is slow. Activated charcoal in particular can cause constipation, and giving a binder to someone who is not moving daily can make the whole picture worse rather than better. It rests on conservative clinical judgment rather than a trial finding, and I searched for supporting literature and did not find it. I use it anyway, because the failure mode it prevents is genuine and the cost of waiting two weeks is zero.

Gate two: an actual documented exposure, and the honest problem with that gate. I do not use binders for a symptom pattern alone. I use them where a mycotoxin panel documents exposure. You should know exactly what that panel is and is not. Published German guidance states plainly that blood or urine tests for mold components or metabolites are not indicated in medical evaluation (PMID 38381662). The commercial panels have not been validated to tell us whether a person is ill from mold or whether the source was a building or the diet, mycotoxins are cleared through urine fast enough that timing dominates the result, and near-universal dietary exposure means a positive cannot localize a source (PMID 33350155, PMID 33924868). The clinical study most often used to support these panels compared results against a non-concurrent, unblinded historical control group and drew two published methodological critiques (PMID 23580077, PMID 27827982, PMID 27827984).

And here is my position, labeled as mine. I order the panel. Not because it is guideline-concordant care, because it is not, and not because a positive result diagnoses anything, because it does not. I order it because I want a documented exposure signal before I put a binder into a reactive patient, and I would rather gate a real intervention on an imperfect test than on

nothing at all. The result never stands alone. It sits next to the exposure history, the building history, and the symptom timeline, and if those three do not agree, the panel loses. The full treatment of this test is in the lab guide for this program.

Activated charcoal, 500 to 1,000 mg, taken 1 to 2 hours away from food, supplements, and medication. The spacing is not arbitrary and it is not 30 minutes. Charcoal adsorbs whatever is in the gut with it, and the effect is strongly time-dependent, with roughly 47 percent absorption reduction at 30 minutes falling to about 17 percent at 120 minutes (PMID 15822758). That number is the reason for the 1 to 2 hour window: too close and it takes your medication and your supplements along with the target. **Evidence: human, for the spacing and adsorption logic. Nothing found for charcoal as a treatment in chemical sensitivity.**

Bentonite or chlorella, at fractional doses. Chlorella deserves a clear-eyed sentence. The binding chemistry is established, chlorophyllin forms complexes with planar polycyclic molecules (PMID 7685489), and a 180-person randomized double-blind placebo-controlled trial of **chlorophyllin** cut a urinary aflatoxin marker by 55 percent (PMID 11724948). That is genuine human randomized evidence, and it used chlorophyllin, not chlorella. The two human chlorella studies are non-randomized, manufacturer-affiliated, and about dioxins in breast milk (PMID 17472477), the animal work used dietary chlorophyll (PMID 11333191), and the metals half of the claim has no human dosing literature at all. **Evidence: tested elsewhere at the class level, animal or test tube only for chlorella itself, nothing found for chlorella and metals.** It is a mechanistically reasonable idea that has not been tested in the form we would actually be using, and fractional dosing is how it goes in.

The practical rules that govern all five sections

One agent every 3 to 5 days, at a quarter to a half of the target dose. This is the rule the rest of the guide is built around. It is slower than you want and it is the only way the log tells you anything.

Hypoallergenic formulations, without unnecessary fillers, dyes, or flavoring. In this population the excipient causes the reaction at least as often as the active ingredient does, and the most common false conclusion I see is a patient who decided they cannot tolerate a nutrient when what they could not tolerate was the capsule. Single-ingredient products beat blends here for the same reason: a blend that provokes a reaction teaches you nothing about which of its nine ingredients did it.

The sequence is fixed. Stabilize and open drainage, then support pathways, then mobilize and bind. Stabilize means sleep, blood sugar, hydration, bowel regularity, and enough exposure reduction that you are not starting a protocol in the middle of a continuous flare. Open drainage means the exit routes are working before anything is asked to leave, with daily bowel movements, adequate fluid, and movement. Only then do the glutathione and cofactor sections go in. Binders come last, and only through both gates in section five. Running this backwards, which is what most aggressive detox programs do, is the single most reliable way to make someone in this population worse.

No supplement in this guide outperforms trigger avoidance. The status of that sentence matters, because it is my clinical position and not a trial result. No study has ever compared avoidance against supplementation in chemical sensitivity, and there is no controlled trial of avoidance at all. Avoidance carries an expert consensus recommendation and is the reasonable clinical default (PMID 34769816). I recommend it first because it is low risk and it directly removes the exposure, not because a trial showed it beats anything else. Everything in this guide exists to raise how much exposure you can absorb before you react. None of it exists to let you stop reducing exposure.

Where to find these, and what I earn

I hold no physical inventory, and I have no way to stock a practitioner-only line. That is a sourcing rule with a practical consequence for you: **where more than one brand would do, I pick a line you can order yourself.** A product that only I can order means I place the order and you pay shipping on top, which is worse for you and worse for the plan, because sourcing friction is one of the most common reasons a protocol quietly stops.

I keep a screened dispensary through Fullscript, which ships directly to you:

us.fullscript.com/welcome/drseay

Where a product is not carried there and it is still the clinically right call, I will name it and you can obtain it from any reputable retailer. Quality matters more than the logo. Look for independent third-party testing, hypoallergenic formulation, and a label that states the actual amount of the active ingredient rather than a proprietary blend total.

Disclosure: Root Health earns a portion of dispensary purchases. Nothing in this guide was included, excluded, ranked, or dosed based on what it earns. Two of the most effective things in this document, the 3 to 5 day pacing rule and reducing your exposures, cost nothing and earn me nothing.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized version of this protocol starts from your labs, your history, your exposure timeline, and your full medication list.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, start with the log. Two weeks of honest tracking before the first capsule is worth more than the first capsule.

Free tracker: rootcauseunlocked.com/tracker

Read this alongside the companion documents in this program. **Unlock Your Chemical Sensitivity** covers the condition, the exposure work, and the nervous system side, and the lab guide covers the testing that decides which parts of this protocol apply to you at all.

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Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

Educational content only. Not medical advice. Not a diagnosis. Never stop, skip, or change a prescription based on this guide. New, severe, or rapidly worsening symptoms need prompt medical evaluation, not a supplement.

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